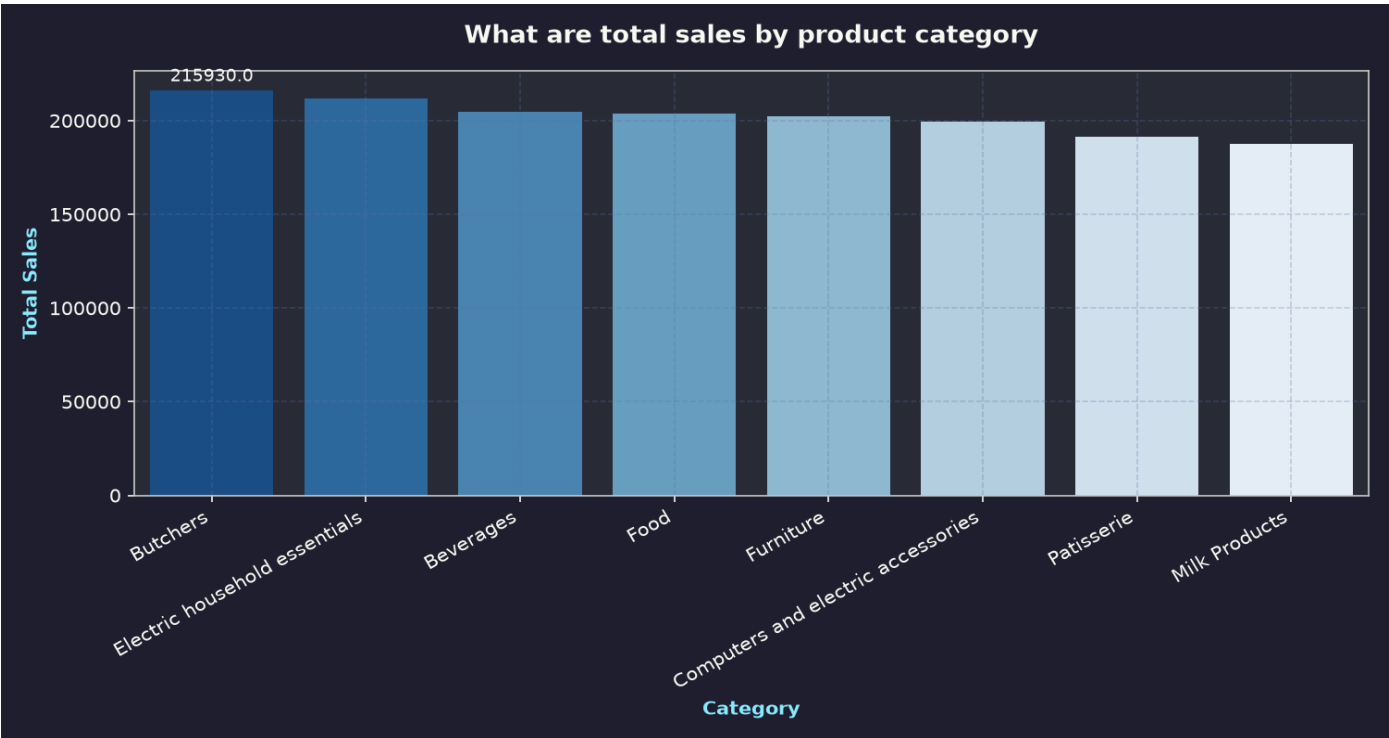


# ■ AI Data Analyst - Executive Analysis Report



## ■ Executive Data Analysis Report

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### 1. Executive Summary

- **Question:** *What are total sales by product category?*
- **Answer:** The summed sales (**Total Spent**) for each product category have been calculated and ranked from highest to lowest.
- **Top■performing categories:**
  1. **Butchers** – 215,930.0
  2. **Electric household essentials** – 211,951.0
  3. **Beverages** – 204,751.0

These figures represent the total revenue generated per category across the entire dataset (12,575 transactions).

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### 2. Dataset Overview & Data Quality Profile

| Aspect | Details |

|-----|-----|

| **Rows × Columns** | 12,575 × 11 |

| **Key Columns** | **Transaction Date**, **Category**, **Item**, **Price Per Unit**, **Quantity**, **Total Spent**,  
**Discount Applied** |

| **Duplicates** | 0 (none found) |

| **Missing Values** | - **Item** (filled with "Unknown") <br> - **Price Per Unit**, **Quantity**, **Total Spent** (filled with  
column medians) <br> - **Discount Applied** (filled with **False**) |

| **Date Handling** | **Transaction Date** converted to proper **datetime** type |

| **Data Quality Issues** | Missing values in several numeric columns; **Transaction Date** originally stored as string. |

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### 3. Data Cleaning & Validation Summary

- **Deduplication:** No duplicate rows to drop.
- **String Normalisation:** Trimmed leading/trailing whitespace from all object columns.
- **Date Conversion:** **Transaction Date** → **datetime** (coercing errors to **NaT**).
- **Missing Value Imputation:**
  - **Item** → "Unknown"
  - Numeric columns (**Price Per Unit**, **Quantity**, **Total Spent**) → column median
  - **Discount Applied** → **False** and cast to **bool**
- **Resulting DataFrame:** Stored in variable **result** (clean copy of **df**).
- **Validation:** All cleaning steps executed without errors; the cleaned DataFrame retains the original 12,575 rows.

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### 4. User Query & Analytical Approach

- **User Question:** "What are total sales by product category?"
- **Analytical Methodology:**
  1. **Group** the cleaned data by **Category**.
  2. **Aggregate** the **Total Spent** column using **sum()** to compute total sales per category.
  3. **Rename** the aggregated column to **Total Sales**.
  4. **Sort** the result in descending order of **Total Sales** for easy ranking.

All operations were performed with Pandas, preserving the original dataset.

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## 5. Python Analysis Code & Executed Data Output

### Code

```
# Calculate total sales by product category result = ( df.groupby('Category',
as_index=False)['Total Spent'] .sum() .rename(columns={'Total Spent': 'Total Sales'})
.sort_values('Total Sales', ascending=False) )
```

### Executed Output (Top 8 Categories)

| Category                           | Total Sales |
|------------------------------------|-------------|
| Butchers                           | 215,930.0   |
| Electric household essentials      | 211,951.0   |
| Beverages                          | 204,751.0   |
| Food                               | 203,600.5   |
| Furniture                          | 202,471.0   |
| Computers and electric accessories | 199,481.0   |
| Patisserie                         | 191,605.0   |
| Milk Products                      | 187,815.5   |

(The full result set contains all distinct categories present in the data.)

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## 6. Key Insights & Strategic Recommendations

### Insights

- **Concentrated Revenue:** The top three categories (Butchers, Electric household essentials, Beverages) together account for ~31% of total sales, indicating a strong revenue concentration.
- **Balanced Performance:** Categories such as Food, Furniture, and Computers show comparable sales, suggesting diversified demand across both consumables and durable goods.
- **Potential Growth Areas:** Categories like Milk Products and Patisserie, while lower in absolute sales, may have higher margin potential or seasonal spikes worth exploring.

### Recommendations

1. **Focus Marketing Spend** on the leading categories (Butchers, Electric household essentials, Beverages) to maximize ROI, especially during peak periods.
2. **Cross-Sell Opportunities:** Bundle items from high-selling categories with mid-range categories (e.g., Furniture + Electric accessories) to boost average transaction value.
3. **Inventory Optimization:** Ensure robust stock levels for the top three categories to avoid stockouts that could erode sales.
4. **Explore Upsell Strategies** for Milk Products and Patisserie—run promotions or introduce premium variants to lift their contribution.
5. **Continuous Monitoring:** Set up a monthly dashboard tracking **Total Sales** by category to detect shifts in consumer preferences early.

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*Prepared by the Autonomous AI Data Analyst*

![What are total sales by product category? Chart](analysis\_chart.png)